

File System Basics

Anyone working with a computer almost always works with a **file system**. It is the organizational system that an operating system uses to manage files and folders on a storage medium—the foundation for everything you do on the computer.



Folder Structure and Hierarchy

The Tree Principle

The folder structure works like a tree: A starting point at the very top, folders beneath it, subfolders inside those, and files at the end. Top = more general, bottom = more specific.

Example

- Main folder: **School**
- Subfolder: **Math**
- File: **Homework.docx**

The file is not just "somewhere," but in a clearly defined place within the structure.

- ❏ Two files can have the same name as long as they are in different folders – e.g., **Documents/English/Notes.txt** and **Documents/History/Notes.txt**.

The Path: Directions in the File System

A **path** describes exactly where a file is located – from the starting point to the specific file.

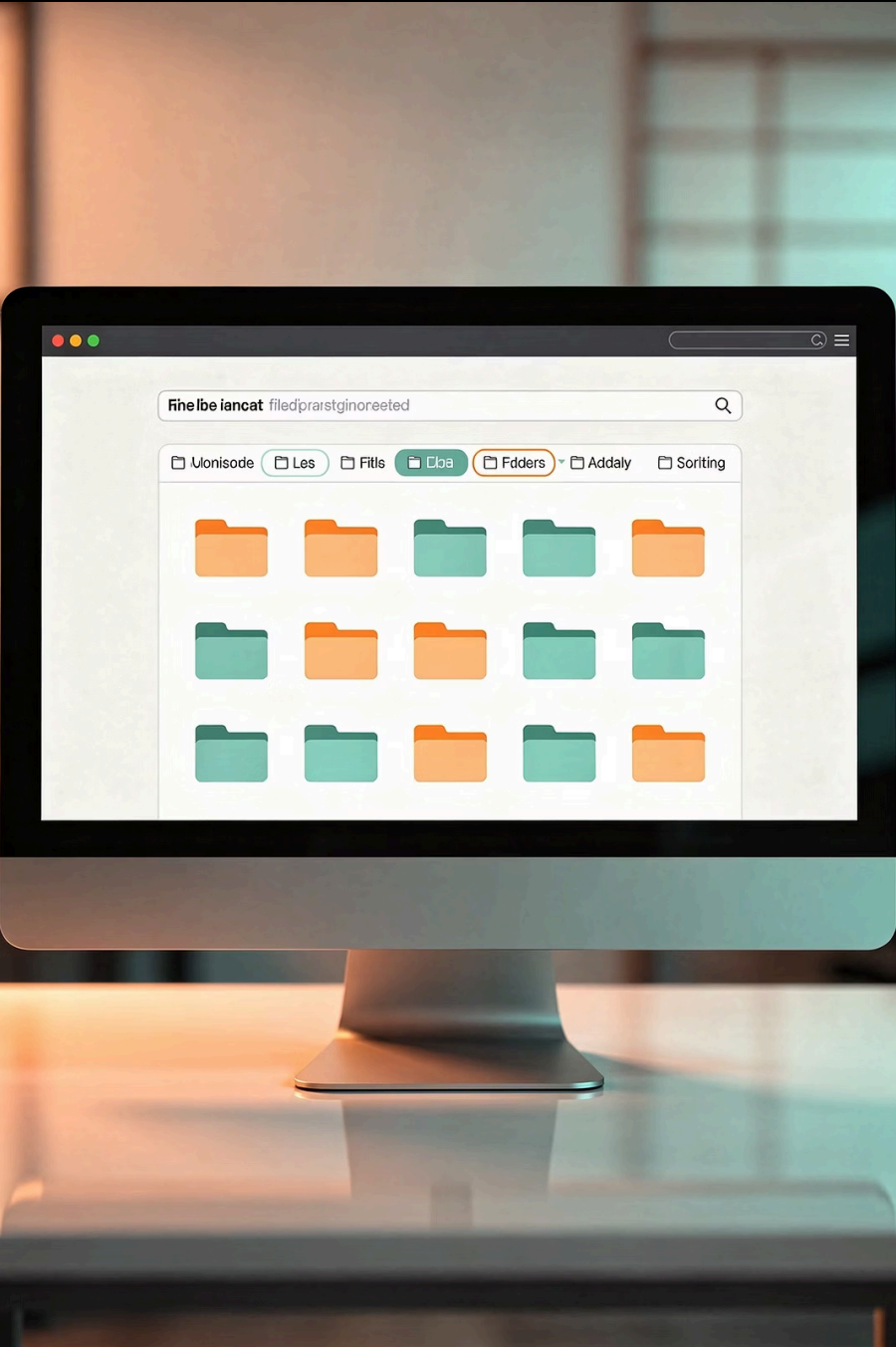
Windows

```
C:\Users\Anna\Documents\School\Math\Homework.docx
```

macOS / Linux

```
/Users/Anna/Documents/School/Math/Homework.docx
```

The notation differs, but the concept is the same: the path shows the hierarchy from the start to the file. On Windows, characters such as `\ / : * ? " < > |` are not allowed in file names.



Why a good folder structure is important



Find faster

Logically stored files can be found again much faster.



Avoid confusion

Clear structures prevent files from being saved twice or in the wrong place.



Work cleanly

Thematically sorted folders – e.g., School, Personal, Applications, Photos – instead of everything on the desktop.

Mindset for Beginners: Three Questions

When you don't know where a file is located, ask yourself these three questions:

1

Main area?

Documents, Downloads, or Pictures?

2

Thematic folder?

School, Work, or Personal?

3

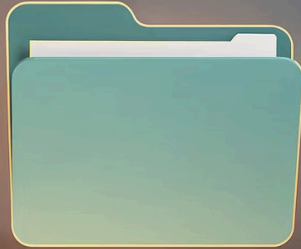
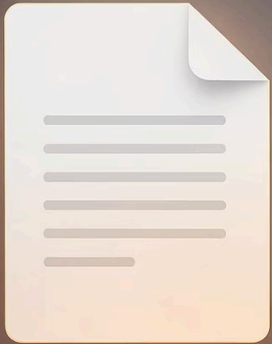
File name?

Presentation.docx, Notes.pdf, or
Invoice_May.pdf?

This is how you learn to see the computer not as a "magic box," but as an organized system—a core area of fundamental digital literacy.

CHAPTER 2

Files and Folders: The Difference



File = Content

A single stored object with content: text document, image, video, PDF, music. Usually has a **file extension** like .docx, .jpg, or .pdf.

Folder = Container

A container that holds files and other subfolders. Usually has **no file extension** and appears with a folder icon.

Files and Folders Compared

Feature	File	Folder
Purpose	Stores content	Organizes content
Can contain	Text, image, audio	Only files/subfolders
Has file extension	Yes (.docx, .jpg ...)	Usually no
Opening behavior	Opens in application	Displays contents
Example	Letter.docx, Photo.jpg	Documents, Pictures

 **Tip:** A file is the actual work object. A folder is the environment in which such objects are collected.

Recognizing File Extensions

The **file extension**—the part after the period in the filename—helps the operating system recognize which file type it is and with what it can be opened.



.docx

Word document



.pdf

PDF file



.jpg

Image



.txt

Text file



.xlsx

Excel spreadsheet

Folders do not have such extensions. A common beginner mistake is to confuse files and folders, because both are displayed as icons.

CHAPTER 3

File Operations: The Six Fundamentals

Anyone who masters these six operations can navigate almost any application and operating system with ease.

1

Create

2

Name

3

Save

4

Copy

5

Move

6

Delete

Creating, Naming, and Saving

Creating vs. Saving

Creating = to generate something new. **Saving** = to permanently secure the current state. Until you save, changes may be lost in the event of a crash.

Good File Naming

- Unique and meaningful
- Short but clear
- Schema: Topic_Date or Subject_Content_Year
- Example: Math_FormulaCollection_2026.pdf
- No special characters like \ / : * ? " < > |

CHECKLIST

Three Things to Check When Saving



Check the file location

Where is the file going? Many beginners click "OK" too quickly—and then the file is "lost".



Check the file name

Is the name clear and unambiguous?
Instead of *Document1.docx*, use something like
Application_Smith_2026.docx.



Check the format

Use "**Save As**" if you want to keep a new version or change the file format.

Copy, Move, and Delete

Copy = Duplicate

An additional version is created in a different location. The **original is preserved**. Ideal for backups or when you want the same file available in multiple places.

Move = Relocate

The file changes location. Afterward, it exists **only at the new location** – like moving a folder from one shelf to another.

Delete = Remove

The file is removed from its storage location. It often goes to the **Recycle Bin** (or Trash) first and can be restored – it is only gone for good once emptied.

Keyboard Shortcuts for File Operations

Windows

Ctrl + C

Copy

Ctrl + X

Cut (to move)

Ctrl + V

Paste

Del

Delete

What happens logically

- **Copy + Paste** → a second version is created
- **Cut + Paste** → object is relocated
- **Delete** → removed from the structure

On Mac, the **Cmd key** is used instead of Ctrl. For beginners, the right-click menu is often clearer because the intention is more explicit.

Common Misconceptions in File Operations

→ **"I moved the file, but it's gone."**

It is not gone – it is simply in a different location.

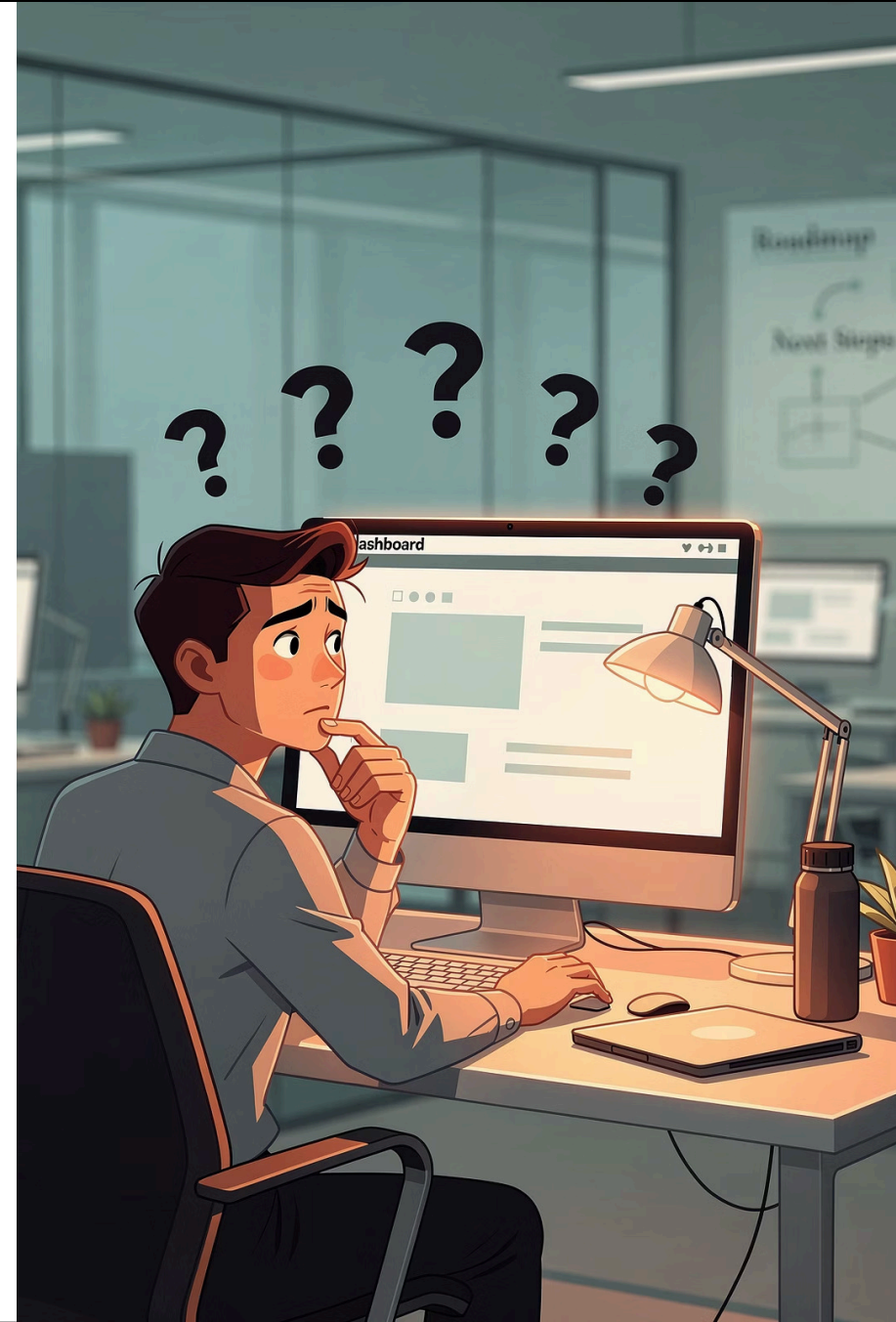
→ **"I copied it, why is everything doubled?"**

Because copying intentionally creates a second version.

→ **"I deleted the folder, but only one file should be gone."**

The container was removed instead of the content – always check what is selected.

📌 **Principle:** Never delete out of habit. Always check *what* is selected first.



CHAPTER 4

Working with Applications

An **application** is a program for specific tasks: writing texts, reading PDFs, viewing images, calculating, printing. Important: Application and file are not the same thing.

Application = Tool

The program you work with.

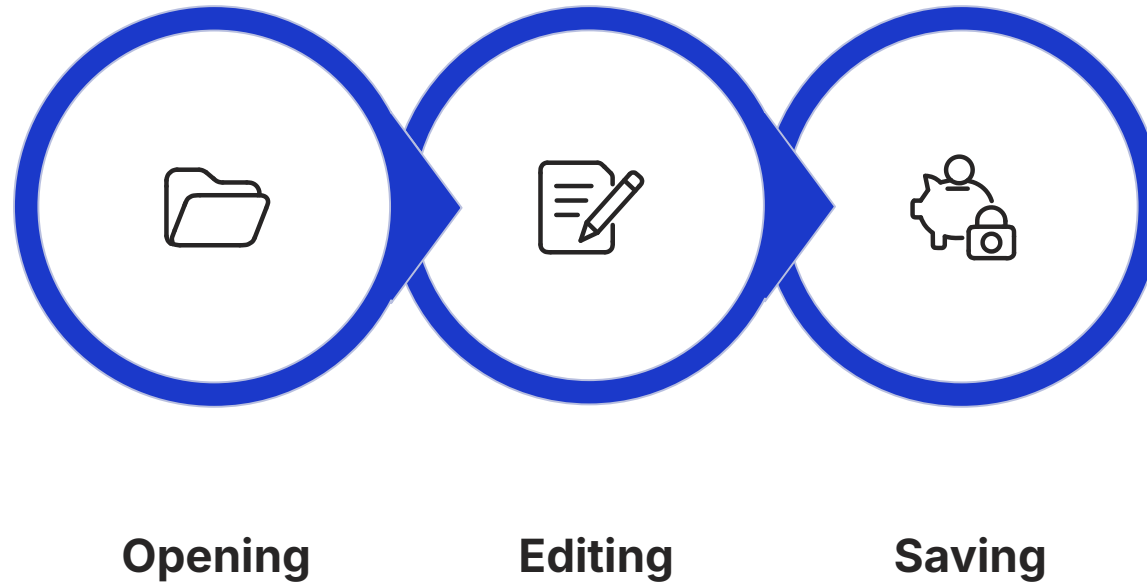


File = Work Product

The result, which is saved in the file system.



Opening, Saving, and Printing Documents



When opening, the file does not change automatically. The file is only actually modified once you edit the content and then save it. Typical file types open automatically in the appropriate application – if necessary, select manually via "**Open with...**".

Save vs. Save As

Standard Save

Updates the existing file. The previous version is overwritten.
Saving early and often is one of the most important habits.

Save As

Creates a new file – with a different name, format, or location.
Useful when you:

- do not want to overwrite a template,
- want to keep different versions,
- want to save a document additionally as a PDF.

Printing Documents: Step-by-Step

01

Open Document

Check the content and save if necessary.

02

Open Print Dialog

File > Print or **Ctrl + P** (Mac: Cmd + P).

03

Check Settings

Printer, page range, number of copies, color/B&W, portrait/landscape orientation.

04

Use Preview

Immediately shows if anything is cut off or if the layout is incorrect.

05

Print

Only then start the print job.

Common situations when opening and saving

Case 1: Changing an existing file

Open the file, edit, save normally – the same file will be updated.

Case 2: Keep the original

Use "**Save As**" and assign a new name.

Case 3: Document cannot be found

The problem usually lies in the file system, not the application – search in the correct folder.

Case 4: File cannot be opened

The format does not match the application or the file is damaged.

CHAPTER 5

Simple Calculations with the Calculator

The calculator is one of the simplest yet most useful applications. It is suitable for quick everyday calculations without needing a spreadsheet program.



Addition

+



Subtraction

-



Multiplication

× or *



Division

÷ or /

Percentage Calculation and Important Keys

Percentage Calculation

Many calculators have a **% key** for typical everyday questions: discounts, sales tax, shares. The safe way to think about it without a special key:

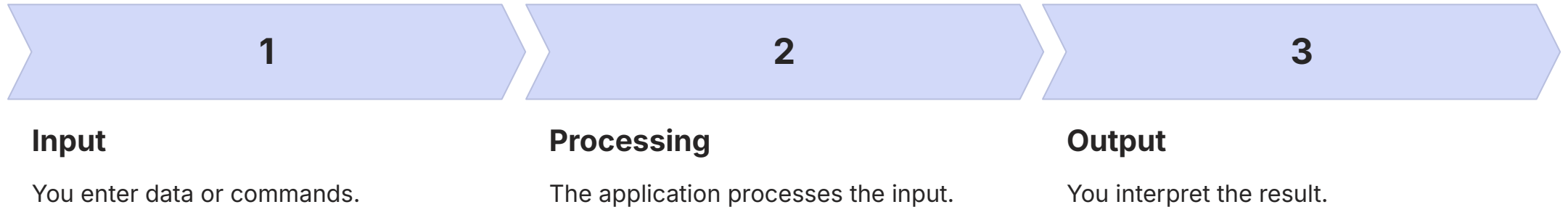
$$200 \times 10 \div 100 = 20 \rightarrow 10\% \text{ of } 200 \text{ is } 20$$

Important Keys

- **C / AC** – Clear entry
- **=** – Show result
- **.** or **,** – Enter decimal number
- **%** – Percentage function

❏ Common errors: choosing the wrong operation or transposing numbers (e.g., 250 instead of 205). Briefly check your input before pressing =.

The Basic Pattern of Every Application



Anyone who understands this pattern with a calculator also understands how other software works in principle. The calculator is therefore more than just a calculating tool – it is an introduction to digital thinking.

CONNECTION

Applications and File System

The four examples show how different applications handle files — from saving to working directly.

Word Processor

Opens and saves files in the file system.



Image Editor

Loads and exports image files.



PDF Reader

Opens, displays, and prints files.



Calculator

Often works directly — without a classic file.

Summary: Digital Literacy



Understanding the File System

Hierarchical folder structure, paths, and the difference between files and folders.



Mastering File Operations

Creating, naming, saving, copying, moving, and deleting—securely and intentionally.



Working with Applications

Opening, editing, saving, printing—and understanding where information resides within the file system.

You are not just learning "which button to press," but understanding **where information resides, how it is organized, and how programs interact with it**. That is the essence of true digital literacy.

